MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Core Course (DSC) - MG3DSCFTQ202 - UNIT OPERATIONS IN FOOD INDUSTRY

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes Maximum Marks: 50

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Part A – Short Answer Questions(Answer any 5 questions, each carries 2 marks)

[2×5 = 10 Marks]

1. Define a scraped surface heat exchanger. [U]/[CO1]

A scraped surface heat exchanger (SSHE) is a type of heat exchanger designed for viscous, sticky, or heat-sensitive food products. It uses rotating blades to continuously scrape the inner surface of the heat exchanger, preventing fouling and ensuring uniform heat transfer.

2. List two heat-sensitive products suitable for freeze drying. [K]/[CO2]

- Coffee extract,Fruit juices or fruit pulps
  (Also applicable: enzymes, probiotics, and vaccines)

3. What are the limitations of water distillation? [U]/[CO1, CO3]

- High energy consumption due to vaporization, Scaling and corrosion of equipment.
- Not suitable for heat-sensitive materials, Low throughput compared to membrane techniques.

4. Why is radiation dominant in high-temperature food processes like broiling? [U]/[CO1]

At high temperatures, heat transfer occurs mainly through electromagnetic radiation rather than conduction or convection because radiant energy can travel directly through air and rapidly heat the food surface.

5. What is regeneration efficiency in plate heat exchangers? [U]/[CO1]

Regeneration efficiency is the ratio of heat recovered from the outgoing (hot) product to the total heat required to heat the incoming (cold) product.

$$\text{Regeneration efficiency} = \frac{\text{Recovered heat}}{\text{Total heat required}} \times 100$$

6. State two advantages of tubular heat exchanger. [U]/[CO1]

- Can handle viscous or particulate foods, Easy to clean and maintain with less fouling. (Also: suitable for high-pressure operations.)

7. Differentiate between freezing and chilling. [U]/[CO1]

Parameter	Chilling	Freezing
Temperature range	Above 0°C	Below 0°C
State of water	Liquid	Solid (ice formation)
Purpose	Short-term preservation	Long-term preservation

[2×5 = 10 Marks]

Part B – Short Essay Questions.(Answer any 5 questions, each carries 4 marks)

8. Recommend nanofiltration for specific food industry applications. [E]/[CO4]

Nano filtration is used for **Dairy industry, Beverage industry, Water purification, Sugar industry**

9. Discuss how direct contact steam injection improves heating efficiency. [A]/[CO3]

- Steam directly contacts the product, condensing and transferring latent heat instantly.
- Minimizes heat loss and provides rapid temperature rise ,Ideal for liquid foods (soups, sauces, milk),Reduces fouling since no heat surface contact.However, steam condensate dilutes the product slightly.

10. Rank the applications of microfiltration in food processing. [E]/[CO4]

Clarification of fruit juices and wines, Removal of bacteria from milk and beer, Separation of fat globules from whey.Pre-treatment for ultrafiltration and reverse osmosis.
Microfiltration enhances microbial safety and product clarity without thermal degradation.

11. Derive the heat conduction equation in cylindrical coordinates and mention its food applications. [U]/[CO1]

$$\frac{\partial T}{\partial t} = \alpha \left(\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial T}{\partial r} \right) \right)$$

Where,

T = temperature, r = radius, t = time, α = thermal diffusivity.

Applications:

- Heat conduction in cans or bottles during sterilization.
- Temperature distribution in cylindrical sausages or pipes.

12. Suggest a suitable drying method for leafy vegetables with reasons. [An]/[CO3]

Recommended method: Freeze drying or vacuum drying.
Reasons: Preserves chlorophyll, color, flavor, and nutrients,Minimizes shrinkage and oxidation.Retains rehydration ability and product quality.

13. Compare the rectifying section and stripping section. [U]/[CO1, CO3]

Section	Function	Composition
Rectifying	Enriches the vapor with more volatile components	High in distillate
Stripping	Removes volatile components from liquid	High in less volatile components
Both are essential parts of a distillation column for separation efficiency.		

14. What are the applications of plate freezers in fish industry? [U]/[CO2]

- Rapid freezing of fish fillets and shrimp in blocks, Uniform freezing with minimal dehydration ,Suitable for packaging in flat cartons or blocks, Maintains texture and freshness for export-quality seafood. **[4×5 = 20 Marks**

Part C – Long Essay Questions(Answer any 2 questions, each carries 10 marks)

15. Compare forward feed and backward feed arrangements in multiple-effect evaporators. [An]/[CO3]

Parameter	Forward Feed	Backward Feed
Flow Direction	Feed enters first effect, flows with steam direction	Feed enters last effect, opposite to steam flow
Suitability	Hot, dilute feeds	Cold, viscous feeds
Steam Economy	Lower	Higher
Equipment Cost	Lower	Higher (requires pumps)
Operation	Easier to control	More complex
Applications: Concentration of fruit juices, dairy products, and sugar syrups.		

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16. List five food products dried by spray drying and explain their importance. [K]/[CO2]

Examples: Milk powder, Coffee powder, Egg powder, Soup mixes – convenience products, Fruit juice powder.
Importance: Preserves flavor, extends shelf life, provides instant rehydration, and reduces transportation cost.

17. Discuss convective mass and heat transfer in drying of fruits and vegetables. [U]/[CO1]

Heat transfer, Mass transfer, Controlling factors

Applications: Hot air drying of mango slices, banana chips, etc. Effective drying ensures energy efficiency and product quality.

18. Produce two examples of industries using RO. [C]/ [CO5]

- **Dairy industry,Beverage industry**
- Other examples: desalination plants, sugar and brewing industries.